

Set 3 (answers p. 222)

1. Most organisms that carry out photosynthesis are able to produce food directly from carbon dioxide and water using energy from light. However, not all organisms use carbon dioxide as a source of carbon atoms for this process. ____ photoheterotrophs use organic compounds rather than carbon dioxide as a source of carbon.

Ⓐ For instance,

Ⓑ Meanwhile,

Ⓒ Furthermore,

Ⓓ Nevertheless,

2. By the beginning of the 20th century, the sonnet had become an antiquated poetic style. Claude McKay, however, found it an ideal medium to convey his ideas. Many modernists criticized his use of the form. ____ he persevered and created a significant number of sonnets.

Ⓐ Moreover,

Ⓑ Likewise,

Ⓒ Indeed,

Ⓓ Nevertheless,

3. The “songs” of blue whales usually register at between 15 and 20 hertz—a nearly inaudible rumble on the edge of what the human ear can perceive. One blue whale in the Pacific has produced vocalizations registering more than 50 hertz, ____ a finding so unusual that some researchers have suggested the animal is actually a hybrid species.

Ⓐ therefore,

Ⓑ moreover,

Ⓒ however,

Ⓓ for example,

4. Growing nutritious, protein-dense microalgae in seawater-fed aquaculture systems—particularly in the southern hemisphere—could help increase food production by more than 50%. ____ researchers estimate that over the next three decades, an additional 10 million people might be fed.

Ⓐ Still,

Ⓑ In fact,

Ⓒ Therefore,

Ⓓ Subsequently,

5. Indo-European languages are the first language of more than 3 billion people in Europe, across northern India, as far east as Siberia, and in North and South America. Beginning almost 500 years ago, scholars began to notice similarities between languages such as ____ it became clear that hundreds of such languages were connected by common root words.

Ⓐ Sanskrit and Latin as the field of linguistics matured,

Ⓑ Sanskrit and Latin, as the field of linguistics matured

Ⓒ Sanskrit and Latin. As the field of linguistics matured,

Ⓓ Sanskrit and Latin as the field of linguistics matured

6. Unlike the binary bits used in standard computers, qubits can exist in multiple states at a time and have much greater processing capacity. However, they require specific conditions to exhibit quantum behavior and are challenging to harness. Synthetic ____ has corrected this problem by creating qubits from the electron spin of molecules with a central metallic atom.

Ⓐ chemist Danna Freedman,

Ⓑ chemist, Danna Freedman,

Ⓒ chemist, Danna Freedman

Ⓓ chemist Danna Freedman

7. It is a well-established fact that exercise can help people lose weight and avoid gaining it. However, identifying the cellular mechanisms that underlie this process ____ difficult because so many cells and tissues are involved.

Ⓐ have proven

Ⓑ has proven

Ⓒ are proving

Ⓓ is proven

8. The algorithms developed by computer scientist Stephanie Gil address such key questions as how robots share information and how each robot enacts an algorithm so that ____ actions are coherent within a group. Her goal is to enable the machines to achieve shared goals more efficiently and accurately.

Ⓐ its

Ⓑ their

Ⓒ they're

Ⓓ our

9. While researching a topic, a student has taken the following notes:

- Gumbo is a heavily seasoned vegetable stew that sometimes includes meat and shellfish.
- Its name comes from a West African word for okra, even though that vegetable is not always used.
- Shrimp and crawfish were introduced by members of the Choctaw tribe.
- Gumbo was most likely invented in the 1700s; the first written references to it appeared in the 19th century.
- Spanish, Caribbean, and German dishes have also influenced gumbo.

The student wants to indicate the diverse origins of gumbo. Which choice most effectively uses relevant information from the notes to achieve this goal?

Ⓐ Named after a West African word for okra, gumbo combines elements of Spanish, Caribbean, German, and Native American cuisine.

Ⓑ Although gumbo was probably invented during the 1700s, it was not mentioned in writing until the following century.

Ⓒ Despite the origins of its name, gumbo does not always include okra.

Ⓓ Some versions of gumbo contain meat and shellfish, whereas others contain only vegetables.

10. While researching a topic, a student has taken the following notes:

- The ATLAS program uses the Large Hadron Collider (LHC) to obtain precise measurements about the formation of the universe.
- The LHC is five stories tall, weighs 7,000 tons, and has more than 1,500 miles of cable.
- More than 5,000 researchers are involved.
- Monica Dunford is the coordinator for research on the Standard Model in physics.
- Silvia Franchini oversees data collection for the project.

The student wants to introduce the scope and purpose of the ATLAS program. Which choice most effectively uses relevant information from the notes to achieve this goal?

Ⓐ The ATLAS experiment was officially funded in 1995; additional countries, universities, and laboratories joined in subsequent years.

Ⓑ Researchers from dozens of institutions on nearly every continent participate in research at ATLAS.

Ⓒ The ATLAS program employs thousands of researchers who use complex tools to investigate the origins of the universe.

Ⓓ As part of the ATLAS program, Monica Dunford coordinates research on the Standard Model, while Silvia Franchini oversees data collection.